Authors: Hyrum Rohrbach, Josi Bartholomew, Josh Hardman, & Paul
Class: Machine Learning 4150
Date: 09/27/26

RouteCast

Abstraction/Objectives

RouteCast is a solution to Utah's most crippling problems: traffic and weather. Often people will go on google maps only to an hour later be stuck in traffic due to unforeseen snowy conditions. This application optimizes planning trips by combining two critical parts of traveling. Firstly RouteCast uses an AI agent to get current traffic and map data from open apis and reasons the fastest route.

Our objectives are to:

- Design three cooperating agents (Weather Agent, Routing Agent, and Combination Agent) with clearly separated responsibilities.
- Produce a final route and estimated travel time that account for both traffic and weather conditions.
- Measure success by accuracy: how closely the estimated time matches the real travel time.

Introduction

Travel time depends on many changing factors. A route that is fastest in clear conditions may be slow or unsafe during a storm. We chose a multi-agent design instead of one large agent so that each agent works with its own data and stays specialized, which keeps data from mixing or overlapping and improves accuracy and precision.

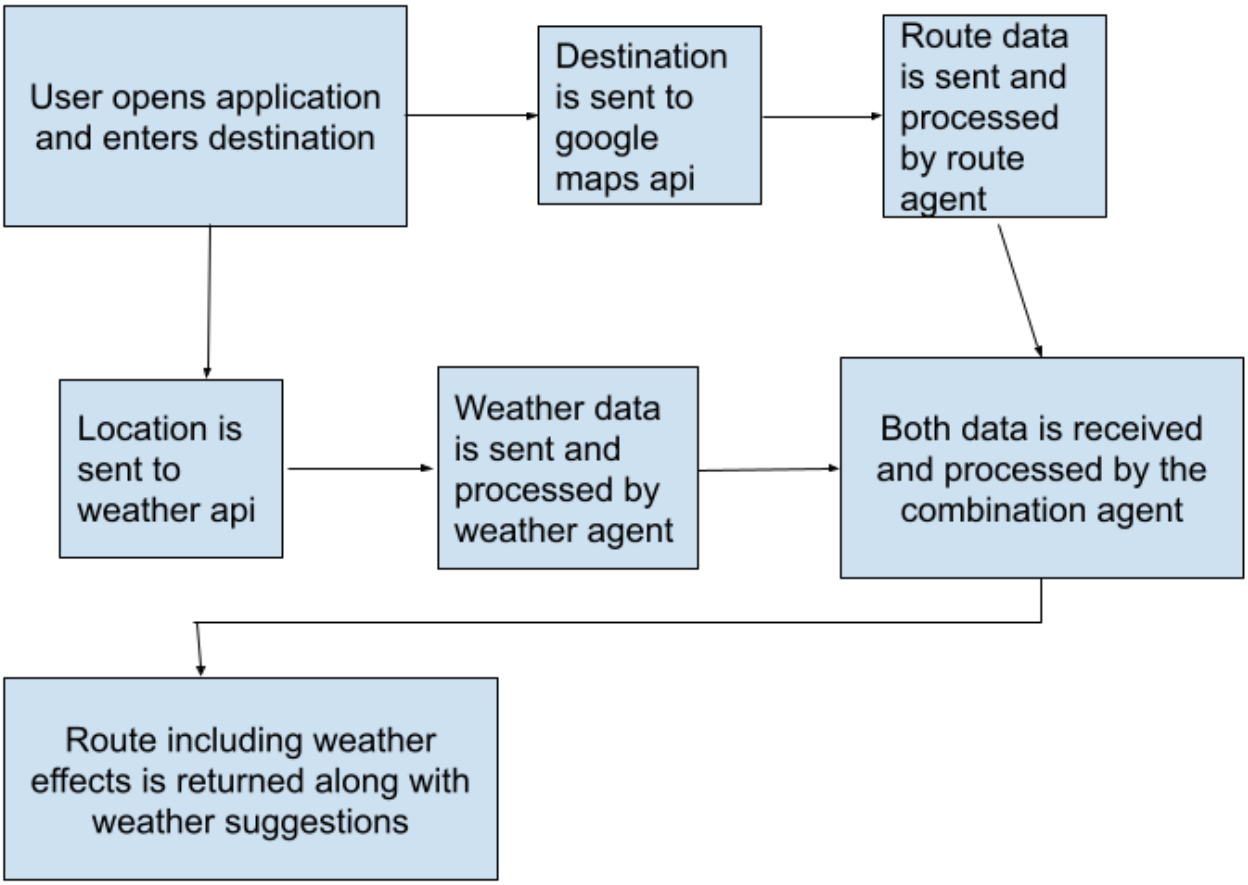
Weather Agent (model-based reflex): It perceives weather data, keeps an internal state of current conditions, and decides how strongly the weather will affect a route, producing a weather-impact score.

Routing Agent (goal-based): It perceives the road network and traffic conditions and plans multiple steps ahead to generate candidate routes from the start to the destination.

Combination Agent (utility-based): It combines the outputs of both agents, weighs trade-offs such as speed against weather risk, and outputs the final route and time estimate. Its PEAS model is: Performance measure – accuracy compared with real travel time; Environment – roads, traffic, and weather conditions; Actuators – the app interface and the displayed travel time; Sensors – the outputs of the Weather and Routing agents.

We will evaluate the system with a performance measure that rewards real-world accuracy rather than raw speed, since an agent optimizing only for speed could behave rationally by that measure and still give poor real-world estimates.

High Level Flow Diagram + Description of the flow diagram



The flow begins when the user opens the application and enters a destination. From this point, the system splits into two parallel paths that run at the same time.

In the first path, the destination is sent to the Google Maps API, which returns route data. That data is passed to the Route Agent, which processes it to generate and evaluate possible routes based on traffic and pathfinding.

In the second path, the user's location is sent to the weather API. The weather data it returns is passed to the Weather Agent, which processes it and determines how the current conditions could affect travel.

Both paths then meet at the Combination Agent. It receives the processed route data from the Route Agent and the processed weather data from the Weather Agent, and it combines them into

a single decision. Finally, the application returns a route that accounts for the weather effects, together with weather-related suggestions for the trip.

Plan of execution
We will build the project in 5 phases.

1. Design: We will define each agent's percepts, actions, and PEAS model and agree on the data passed between agents
2. Data: We will select weather, traffic, and map sources and build the data ingestion.
3. Agent Development: we implement the Weather Agent, the Routing Agent (with a pathfinding algorithm), and the Combination Agent's utility function.
4. Integration and Testing: We will connect the agents to the interface and test normal and failure cases, such as a weather source going offline, in which case a backup weather source would take over.
5. Evaluation: We will compare estimates against real travel times, tune the utility weights, and finish the final report and demo.

Weekly Plan of Action and work load distribution

Week	Dates	Tasks	Owner(s)
1	Sep 28 – Oct 4	Finalize requirements, agent interfaces, and PEAS models; choose data sources and set up a shared repository.	All
2	Oct 5 – Oct 11	Weather data ingestion; road network and traffic data ingestion; UI wireframes.	Josh; Paul; Josi
3	Oct 12 – Oct 18	Weather Agent internal state and weather-impact scoring; baseline pathfinding in Routing Agent.	Josh; Paul
4	Oct 19 – Oct 25	Traffic-weighted route costs; Combination Agent utility function first draft; UI skeleton.	Paul; Hyrum; Josi
5	Oct 26 – Nov 1	Integrate all three agents end to end (checkpoint demo).	All
6	Nov 2 – Nov 8	Tune utility weights (time vs. weather risk); connect UI to the Combination output.	Hyrum; Josi
7	Nov 9 – Nov 15	Sensor-failure handling and backup weather source; unit and integration testing.	Josh; Josi
8	Nov 16 – Nov 22	Evaluate ETA accuracy against real travel times; analyze and fix errors.	Hyrum; Paul
9	Nov 23 – Nov 29	Bug fixes and documentation (Thanksgiving week, lighter workload).	All
10	Nov 30 – Dec 6	Final report, demo preparation, and submission.	All